LUCAS MOUGEOT

Systems Engineer | Distributed Platforms, Robotics Integration & Applied AI Infrastructure

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SUMMARY

Systems engineer designing distributed infrastructure and platform services integrating robotics, AI inference, and cloud-native deployments. Experienced designing deterministic, auditable systems spanning autonomy integration, backend services, and cloud-native deployments. Strong background in Python, C++, and TypeScript with hands-on experience in Dockerized microservices, CI/CD, infrastructure-as-code, and platform tooling. Known for building reliable systems that operate in regulated and real-world environments.

EXPERIENCE

Integration Engineer — EZ Systems (Full-Time) Feb 2025 – Aug 2025

- Designed production infrastructure, environment configuration management, and CI/CD workflows supporting AI inference, sensor pipelines, and robotic deployments with versioned releases, rollback safety, and reproducible environments.
- Implemented cross-system integration tooling coordinating robotics firmware, cloud services, and edge compute nodes.
- Built a real-time ADA compliance auditor integrating computer vision, LiDAR-derived measurements, and regulatory parsing, supported by a unified QA and controlled ML deployment framework with versioned model tracking.
- Developed robotic scan-path generation software and defined deployment architecture for early pilot of autonomous ground vehicle in solar-field environments.

Founding Systems Engineer — AI-Driven Compliance Platform 2024 – Present

- Architected a distributed, production-grade AI-assisted platform with versioned workflows, strict service boundaries, and infrastructure tooling supporting scalable ingestion and deterministic system behavior.
- Deployed multi-service Docker architectures using PostgreSQL (Supabase), Redis, and BullMQ workers with strict API / worker separation.
- Implemented structured logging, observability instrumentation, and comprehensive test suites (fuzz, concurrency, regression, E2E) to ensure production reliability and safe iteration under change.
- Designed cloud-hosted backend services and schema-driven intake systems powering versioned workflows, validation, and analytics-ready datasets.
- Built versioned legal knowledge graphs using PostgreSQL ltree with temporal validity windows, overrides, and cryptographically chained audit logs.
- Implemented hybrid RAG pipelines (vector + full-text) with citation-anchored synthesis and hallucination filtering.

Graduate Researcher — Materials Science (NASA-Funded Capillary Devices) 2023 – 2024

- Designed and fabricated porous ceramic devices enabling passive capillary transport in microgravity environments for space agriculture.
- Designed reproducible experimental data workflows and structured evaluation pipelines to ensure traceability and scientific integrity.
- Developed 36 ceramic formulations with controlled pore structure; executed iterative experiments to optimize capillary rise, stability, and manufacturability.
- Built MATLAB computer-vision pipelines for pore segmentation, morphology quantification, and statistical ranking of candidate formulations.
- Characterized microstructure and composition using SEM/EDS; documented processing–structure–property relationships.
- Authored SOPs, DOEs, and failure-mode documentation; delivered a top formulation achieving ~3× NASA baseline performance in bench tests.

Undergraduate Researcher — Computational Geometry (NSF-Funded) 2020 – 2022

- Developed and implemented the degree-4 plane geometric spanner in modern C++ using CGAL under strict planarity and stretch-factor constraints.
- Developed point categorization and classification algorithms guiding spanner construction decisions based on geometric and neighborhood properties.

- Work contributed to a peer-reviewed ACM Journal of Experimental Algorithmics publication and SoCG media exposition.

SELECTED PROJECTS

Automated Lithium-Ion Battery Pack Assembler Senior Design Capstone

- Designed and implemented an automated lithium-ion battery pack assembly system using an AutomationDirect Do-More PLC.
- Developed ladder logic coordinating motion sequencing, interlocks, fault handling, and safe state transitions.
- Integrated an RP Series robotic dispenser via RS-232 serial communication to initiate external job file execution.
- Implemented PWM-based servo motor control for precise positioning and timing.
- Delivered a deterministic, repeatable automation pipeline with full I/O mapping and bring-up procedures prior to generative AI tools.

EDUCATION

University of North Florida

M.S. Materials Science & Engineering (Thesis-Based) — GPA: 4.00 / 4.00

- Thesis: Development and Characterization of Ceramic Devices for Capillary Action in Microgravity for the Purpose of Growing Vegetables in Space (NASA-Funded)

B.S. Electrical Engineering — GPA: 3.93 / 4.00

B.A. Interdisciplinary Studies: Morphology of Futurism — GPA: 4.00 / 4.00

Concentrations: Mechanical Engineering, Materials Science, Chemistry, Physics, Computing, Mathematics, Mandarin Chinese

PUBLICATIONS & PRESENTATIONS

- Bounded-Degree Plane Geometric Spanners in Practice — ACM Journal of Experimental Algorithmics (Peer-Reviewed)
- Highly Sensitive and Stable pH Sensor Electrodes of TiN Fabricated Using HiPIMS with Kick — AVS 69 Poster, 2023
- Interactive Tool for Bounded-Degree Plane Geometric Spanners — SoCG Media Exposition, 2021

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL
Infrastructure & Cloud: Docker, PostgreSQL, Redis, AWS, Terraform, GitHub Actions (CI/CD), Service-Oriented Architectures
Distributed Systems: Background Workers (BullMQ), API / Worker separation, Versioned Data Models, Hybrid RAG Pipelines
Observability & Reliability: Structured Logging, Regression Testing, Concurrency Testing
Robotics & Autonomy: Motion Planning, Scan Path Generation, PLC Automation, Sensor Integration